Where Does Heterogeneous Federated Learning Lose Rare Classes? A Stage-by-Stage Comparison with Centralised Training

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Paper under double-blind review

Abstract

Generator-based model-heterogeneous federated learning lets clients with different architectures share knowledge through a frozen feature extractor, a conditional feature generator trained by federated averaging, and architecture-specific heads. It has been evaluated only on balanced clients. We ask where, under long-tailed label skew, such a pipeline loses accuracy relative to the *same* pipeline trained on the pooled data. Since a client partition is a pure label shift, federation cannot bias class-conditional targets; it loses accuracy only through the optimiser, through parameters tied to a client's label prior, and through sample size. (i) In the generator, Adam's coupled weight decay on clients that lack a class shrinks that class's conditioning parameters at Adam's step size; a ratio computed from class counts alone predicts the collapse (all 352 of 900 rows below threshold collapsed; 16 of 16 in a prospective test). (ii) The largest loss, 7–20 points of balanced accuracy, is the last layer; a readout from securely aggregated class statistics equals its centralised counterpart, whereas per-client logit corrections are exact reparametrisations that cannot change the representation. (iii) Synthetic features help the representation only if they carry their label: label-independent synthetic data penalise the spread of the logits, and the gain of the synthetic phase changes sign with the generator's signal fraction. Repairing each stage brings the pipeline within 0–2 points of centralised training where we could measure it, but on fresh seeds its margin over the strongest of nine baselines is only -0.3 to $+1.9$ points. The collapse mechanism also removes rare item embeddings in federated matrix factorisation.

1 Introduction

Federated deployments rarely have one model architecture or balanced labels: sites run different networks on different hardware, and rare classes are seen by few of them. Generator-based model-heterogeneous federated learning (GeFL and its feature-space variant GeFL-F [1]) addresses the first problem. Parameters of different architectures cannot be averaged, so the clients instead train a shared conditional generator by FedAvg [2], and every architecture learns from its samples. Its evaluation used IID clients.

We study the second problem in this setting and ask a diagnostic question: *when labels are long-tailed across clients, how far is the federated solution from the centralised one, and which stage is responsible?* We compare each stage of GeFL-F with its counterpart trained on the pooled data, with proofs where the question has an exact answer and measurements where it does not. Our findings are:

1. **Federation adds no target bias** (Section 2): the partition is a pure label shift, and the extractor is, to first order, trained on the pooled objective.
2. **The generator loses rare classes through the optimiser** (Section 3). With Adam and coupled weight decay, a client that lacks a class moves that class's conditioning row toward zero at Adam's full step size, while holders barely grow it. A count-only step ratio κ_c predicts which rows collapse, retrospectively and prospectively (Fig. 1). Decoupled decay or averaging each row over its holders removes the collapse.

3. **The last layer is the largest loss and is recovered exactly** (Section 4). Per-client logit corrections (logit adjustment, FedLC, FedRS, balanced softmax) are reparametrisations of a single-client head; a readout from aggregated class statistics equals the centralised readout and recovers 7–20 points.
4. **Synthetic data help the representation only if they carry their label** (Section 4.3). Label-independent synthetic data penalise the spread of the logits; the gain of the synthetic phase changes sign with the generator's signal fraction (Fig. 2).

We then repair each stage with exactly aggregated statistics (Section 5) and test the result (Section 6). Where we could build the centralised comparator (MNIST, FashionMNIST, CIFAR-100), the federated pipeline loses about as much to the long tail as centralised training does. Against nine baselines on fresh seeds with validation-selected settings, however, its margin is small (-0.3 to $+1.9$ points): removing the coupled decay and adding the exact readout recover most of the loss, which is itself a finding. Finally, the generator mechanism is an instance of a general one: in federated matrix factorisation on MovieLens-1M, rare item embeddings collapse in the same way (Section 6.5).

Contributions. (1) A stage-by-stage decomposition of the federated–centralised gap, assembled from short exact results (the tools are classical; the assembly is new), each checked against a measurement. (2) A count-only predictor of rare-class collapse in FedAvg-trained conditional models, confirmed retrospectively, prospectively and by intervention. (3) A measured ranking of the losses: last layer 7–20 points, generator collapse 7–11, one-client representations 3–4, label-free synthetic data up to 2, extractor 0–1. (4) A pre-registered evaluation with honest margins, including the predictions that failed.

2 Setting and the centralised comparator

GeFL-F. There are K clients; client k holds D_k and trains a head of architecture $a(k)$ out of G architectures. (1) A small feature extractor (FE) ϕ is trained with the heads by FedAvg and frozen. (2) A conditional VAE $G_F(h \mid y)$ is trained on features $h = \phi(x)$ by FedAvg, with Adam (learning rate 10^{-3}) and *coupled* weight decay $\lambda = 10^{-3}$, as in [1]. The label enters through a linear projection whose c-th column, the *conditioning row* $r_c \in \mathbb{R}^d$, is the only class-specific parameter. (3) Each round, each client trains its head for T_s epochs on generated features with uniformly drawn labels and then for $T_r = 5$ epochs on its real features; heads of the same architecture are averaged.

Notation. n_{kc} is client k's count of class c, $n_c = \sum_k n_{kc}$, $N = \sum_c n_c$, $\pi_{kc} = n_{kc}/n_k$. H_c is the set of m_c clients holding c. A long-tailed split draws class sizes $n_c \propto \mathrm{IF}^{-c/(C-1)}$, randomly permutes each class and cuts it into Dirichlet shares.

Lemma 1 (pure label shift)**.** *For every client k and class c with $n_{kc} > 0$, client k's class-c samples are a uniformly random subset of the pooled class-c samples. Hence $P_k(x \mid y) = P(x \mid y)$, and the same holds for $h = \phi(x)$.*

So for any class-conditional quantity (a class row, a class mean, a readout) every client's data term points at the centralised target. Federation can cost *weighting, variance and rate*, but a *bias* can only come from (i) the optimiser, (ii) parameters coupled to a client's label prior (biases, last layers), or (iii) shared parameters trained on prior-weighted losses, which are equally prior-weighted centrally. Sections 3 and 4 locate each.

The centralised comparator. We compare against GeFL-F's own two-stage design trained on the pooled data: an extractor (federated, or central with balanced sampling), then each architecture's body trained on the pooled real features with the same schedule, then the same class-balanced readout. This isolates what federation costs given the frozen-extractor design. The metric is balanced accuracy.

Stage I is first-order pooled training. FedAvg with $p_k = 1/K$ and $\tau_k \propto n_k$ local steps has, to first order, the surrogate objective $\sum_k \frac{p_k \tau_k}{\bar{\tau}} \mathscr{L}_k$ [35], whose weights on our splits equal n_k/N up to a total variation of 0.009–0.018 at $K = 10$ (Appendix A). The regime is not perturbative [36, 37], so we test it directly: replacing the federated extractor by a centrally trained, class-balanced one changes the accuracy of a pooled body with the exact readout by 0.0, $+1.1$ and -1.0 points (MNIST, FashionMNIST, CIFAR-100; Section 6.3).

3 Stage II: why the generator loses rare classes

3.1 A contraction bound and the role of Adam

In one round, client k starts from the global row r_c, runs local Adam steps, and returns $r_{k,c}$. Let $\Delta_k = \|r_{k,c}\| - \|r_c\|$, the holders' mean gain $\bar{G}_c = \frac{1}{m_c} \sum_{k \in H_c} \Delta_k$ and the non-holders' mean loss $\bar{D}_c = -\frac{1}{K - m_c} \sum_{k \notin H_c} \Delta_k$; $p_c = m_c/K$.

Theorem 2 (contraction bound)**.** *Under flat averaging $r_c^+ = \frac{1}{K} \sum_k r_{k,c}$, $\|r_c^+\| \le \|r_c\| + p_c \bar{G}_c - (1 - p_c)\bar{D}_c$, with equality iff all $r_{k,c}$ are parallel. The row contracts whenever $p_c < \theta_c := \bar{D}_c/(\bar{G}_c + \bar{D}_c)$.*

The bound is the triangle inequality applied to measured changes; it describes, it does not predict. What the optimiser decides is the size of $\bar{D}_c$:

Lemma 3 (coupled decay under Adam)**.** *A non-holder's gradient on r_c is exactly λr_c. Adam is invariant to a positive rescaling of the gradient sequence up to ε, so while $\lambda |r_{c,j}| \gg \varepsilon$ each non-holder step moves every coordinate of r_c toward zero by about the learning rate η, independently of λ. At $\lambda = 0$, $\bar{D}_c = 0$ exactly. Under SGD the same decay costs only $\approx S\eta\lambda \|r_c\|$ per round.*

A holder's data gradient fits the row and need not grow its norm, so $\bar{G}_c \ll \bar{D}_c$, $\theta_c > \frac{1}{2}$, and *rare-class rows contract even when most clients hold the class*. On an instrumented run (MNIST, IF $= 100$, $K = 10$) every class is held by at least 7 of 10 clients, yet the rows of the three rarest classes shrank to 4–10% of their initial norm while head rows grew by $1.4\times$. $\lambda = 10^{-4}$ and 10^{-3} gave the same trajectories (final tail/head ratio 0.155 vs 0.152); at $\lambda = 0$ it stayed at 0.85. Since even a class held by every client can contract when holder updates disagree in direction, no threshold on m_c alone can be correct.

3.2 A count-only predictor

Approximation 4 (step ratio)**.** Per coordinate an Adam step is $\eta\psi$ with $\psi = \hat{m}/\sqrt{\hat{v}} \in [-1, 1]$. A non-holder sees a constant-sign decay gradient, so $\psi = \operatorname{sign}(r_{cj})$ on each of its τ_k steps; a holder moves the row only on its $\tau_k f_{kc}$ steps whose batch contains c (f_{kc} is a hypergeometric probability), with $|\psi| \le 1$. The drift per round under flat averaging is therefore

$$\Delta r_{cj} = \frac{\eta}{K} \Big[\psi_h \sum_{k \in H_c} \tau_k f_{kc} - \operatorname{sign}(r_{cj}) \sum_{k \notin H_c} \tau_k \Big], \qquad \kappa_c = \frac{\sum_{k \in H_c} \tau_k f_{kc}}{\sum_{k \notin H_c} \tau_k}. \tag{1}$$

If $\kappa_c < 1$ the drift points toward zero for every coordinate and every target: the row collapses up to an $O(\eta\tau/K)$ oscillation. If $\kappa_c > 1$ it settles where the holders' normalised step is $1/\kappa_c$.

This neglects momentum carried between sparse holder steps and the drift of $\hat{v}$. κ_c uses only the class counts and the number of local steps, both known before training.

Figure 1: $\kappa_c < 1$ has been sufficient for collapse in all 368 rows; $\kappa_c \ge 1$ is not sufficient to survive when the holders' signal is weak. Instrumenting every round, the per-round shrinkage on non-holders is 2–10 times the holders' growth under coupled decay and about 10^{-3} or less with decoupled decay.

Why centralised Adam does not collapse these rows. Centrally, $\hat{v}$ for r_c is dominated by data gradients, so decay steps are $\lambda r/\sqrt{\hat{v}} \ll \eta$, provided the row receives data more often than Adam's

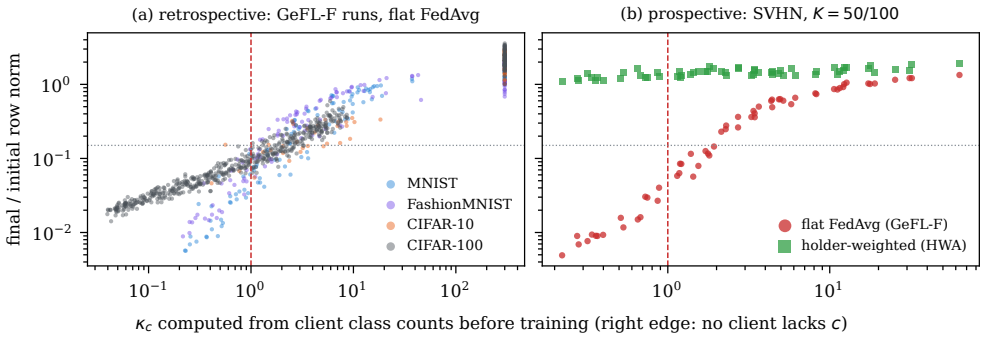


Figure 1: Class counts alone predict the collapse of conditioning rows. Final over initial row norm against κ_c (Eq. (1)); dashed: $\kappa_c = 1$. (a) All 900 logged rows of GeFL-F runs (MNIST, FashionMNIST at $K = 10, 50, 100$; CIFAR-10; CIFAR-100): all 352 rows with $\kappa_c < 1$ end at ≤ 0.153 of their initial norm; Spearman 0.94. (b) Prediction fixed before the runs: all 16 rows with $\kappa_c < 1$ collapse under flat averaging (largest norm 0.04), none under holder-weighted averaging (smallest 1.25).

Table 1: Fixing the collapse (balanced accuracy, long tail, 3 seeds, heads with LA). Every fix restores the rare rows (tail/head norm 0.73–0.97 against 0.04–0.34).

	MNIST			FashionMNIST		
K	10	50	100	10	50	100
GeFL-F (coupled decay)	79.2	81.4	79.5	65.2	66.5	66.6
+ AdamW (decoupled decay)	86.4	84.7	81.6	75.9	74.0	71.8
+ no decay on class rows only	86.8	84.4	81.8	75.9	74.5	71.9
+ HWA (coupled decay kept)	**87.9**	**86.5**	**83.1**	76.0	**75.5**	**73.3**
HWA − best optimiser fix	+1.1	+1.8	+1.3	−0.1	+0.9	+1.4

second moment forgets ($1/(1-\beta_2) = 1{,}000$ steps), as every class does here. In federation a non-holder's $\hat{v}$ contains only decay. When the frequency condition fails centrally too, as for rare items in recommendation, centralised training collapses as well (Section 6.5).

3.3 Fixes: the optimiser or the aggregation

If the coupled decay on non-holders causes the collapse, any fix that removes it should restore the rows. *Holder-weighted aggregation* (HWA) averages each row only over the clients that hold the class, with weights $E(n_{kc}) = (1-\beta^{n_{kc}})/(1-\beta)$ [25], $\beta = 0.999$: $r_c \leftarrow \sum_k E(n_{kc}) r_{k,c} / \sum_k E(n_{kc})$; it needs only two sums, so it runs under secure aggregation [38]. It removes the $\bar{D}_c$ term of Theorem 2 and, without decay, also the m_c/K dilution of a rare row's learning rate (Proposition 9 in the appendix).

Table 1: removing the coupled decay recovers 2–11 points; removing it only on the class rows is as good as removing it globally, so the decay *on the rows* is the cause. HWA adds 1.3–1.8 points at $K = 50, 100$ on MNIST, the regime where the dilution of a rare row's rate is largest (the HWA/flat rate ratio of MNIST's rarest class is 1.15, 2.28 and 3.14 at $K = 10, 50, 100$).

4 Stage III: heads, readout and synthetic data

4.1 Per-client logit corrections are reparametrisations

Proposition 5. *Let a head have a free last-layer bias b, trained by SGD without decay on b, and let $a_k \in \mathbb{R}^C$ be a client constant: logit adjustment $a_k = \tau \log \pi_k$ [26], FedLC's calibration [16] or balanced softmax. (1) Training with $\ell(f + a_k, y)$ from (θ_0, b_0) produces exactly the trajectory of training with $\ell(f, y)$ from $(\theta_0, b_0 + a_k)$; every parameter except b is identical at every step. (2) FedRS [17], which scales absent-class logits by α, is plain cross-entropy in rescaled variables*

$\tilde{w}_c = \alpha w_c$, $\tilde{b}_c = \alpha b_c$ with the learning rate on those rows multiplied by α^2; the body follows the same trajectory.

For a single-client head trained on real data, these four published corrections therefore act only through the readout. Measured on five datasets and three fresh seeds, real-data heads trained with cross-entropy, LA, FedLC or FedRS reach the same exact-readout accuracy within 0.5 points on every seed and dataset. The identity breaks in two places that matter here: when clients with different a_k are averaged, and when a synthetic phase on unadjusted logits alternates with an adjusted real phase (Section 4.3).

4.2 The readout is partition-invariant

Theorem 6 (partition invariance)**.** *Let an estimator be a function $\Psi(S)$ of $S = \sum_k \sum_{i \in D_k} T(x_i, y_i)$ with T fixed in advance. Its value is the same for every partition of the pooled data and equals the centralised estimator.*

The theorem is trivial; its use is to sort the pipeline. The *exact readout* (ER) replaces each architecture's last layer by the class-balanced linear discriminant computed from per-class counts, sums $\sum e$ and second moments $\sum ee^\top$ of that architecture's embeddings e over *all* clients (one extra round): with class means μ_c and pooled within-class covariance S shrunk to $S_\gamma = (1-\gamma)S + \gamma \frac{\operatorname{tr} S}{d} I$, $\gamma = 0.1$, it sets $w_c = S_\gamma^{-1}\mu_c$, $b_c = -\frac{1}{2}\mu_c^\top S_\gamma^{-1}\mu_c$. By Theorem 6 ER, Fed3R [22], FedNCM [23] and all-client CCVR [14] equal their centralised values for the same bodies. The extractor, generator and bodies trained by FedAvg are not partition-invariant; neither are LA, FedLC and FedRS, which are client-specific by design. Two consequences: the federated last layer stops being a source of loss relative to centralised training of the same bodies; and ER accuracy measures the body, not its last layer. A head trained on one client is the extreme case of minority collapse [33]: absent classes have imbalance ratio ∞, and ER sidesteps it.

4.3 Alternation and the signal fraction

Each round a client runs S_s synthetic steps, then S_r real steps.

Proposition 7 (alternation; quadratic losses)**.** *Let the synthetic and real losses be quadratic with Hessians $H_s, H_r \succeq 0$ and minimisers θ_s, θ_r; $P_s = (I - \eta H_s)^{S_s}$, $P_r = (I - \eta H_r)^{S_r}$. If $\|P_r P_s\| < 1$ the round map converges to $\theta^\star = \theta_r + \Omega(\theta_s - \theta_r)$, $\Omega = (I - P_r P_s)^{-1} P_r (I - P_s)$. For commuting Hessians the synthetic weight along eigendirection i is $\omega_i = \rho_{r,i}(1-\rho_{s,i})/(1-\rho_{r,i}\rho_{s,i}) \le \rho_{r,i}$.*

Along directions the client's real loss leaves free ($h_{r,i} = 0$) the synthetic data decide; these are exactly the directions of the classes the client lacks. This gives logit adjustment a second role during training: without it the real phase pushes absent-class logits down without bound, as on separable data [34], and erases what the synthetic phase taught. Measured on the real pipeline, the fraction of the synthetic phase's change to absent-class rows that survives the next real phase is 0.96–1.0 with LA and 0.76–0.98 without.

What the synthetic phase teaches depends on whether its samples carry their label.

Proposition 8 (label-free synthetic data penalise logit spread)**.** *If the synthetic label y is uniform on $[C]$ and independent of the generated feature, then for every logit vector f*

$$\mathbb{E}_y\, \ell(f,y) = \operatorname{LSE}(f) - \bar{f} = \log C + \operatorname{KL}\big(u \,\|\, \operatorname{softmax} f\big) = \log C + \tfrac{1}{2C}\textstyle\sum_c (f_c - \bar{f})^2 + O(\|f - \bar{f}\|^3).$$

If a generated sample carries its label's information with probability s (the signal fraction*) and is label-independent otherwise, the synthetic loss is $s\,\mathbb{E}[\ell \mid \textit{faithful}] + (1-s)\,\mathbb{E}\operatorname{KL}(u \,\|\, p_f)$.*

At $s \approx 0$ the synthetic phase is an explicit anti-discrimination regulariser on the generator's support, and by Proposition 7 it survives exactly in the absent-class directions. The prediction is a sign change: synthetic data hurt the body when s is near zero and help once the faithful term dominates. Figure 2 confirms it. We estimate $s = (a - 1/C)/(1 - 1/C)$ from the accuracy a of a

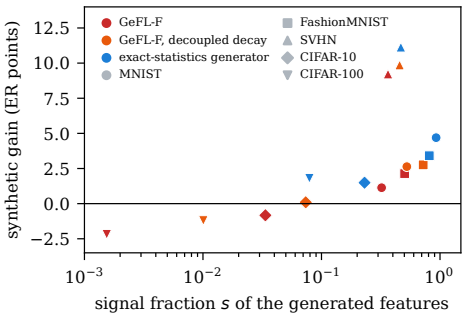


Figure 2: The synthetic phase helps the representation only when generated features carry their label. Gain of the synthetic phase (exact-readout accuracy minus that of heads trained on real data only, same seeds and splits) against the signal fraction s of the generated features; mean of three fresh seeds per point. The gain changes sign between $s \approx 0.03$ and 0.07 (CIFAR-10, CIFAR-100). SVHN gains most at moderate s because its real-only bodies are the weakest.

referee classifier, trained on held-out real features, on generated features. On CIFAR-100, GeFL-F's generator ($s = 0.002$) makes the bodies 2.2 points worse than training on real data alone; the decay-fixed generator ($s = 0.010$) 1.2 points worse. Within every dataset the gain increases with s. On the real pipeline, the spread of the logits on generated features after the synthetic phase relative to before it settles at 0.5–0.6 for GeFL-F and 0.7–0.8 for the higher-s generators, as Proposition 8 predicts.

5 Repairing each stage with exact statistics

The diagnosis suggests one principle: wherever the data decide the result, use a statistic that secure aggregation computes exactly (a sum over clients of per-class quantities) instead of a parameter that FedAvg estimates. Every statistic below costs one upload per client and reveals only federation-level sums. The principle is not new as a readout (CCVR, Fed3R, FedNCM); we apply it to the generator.

Generator. (i) *Mean anchor.* The decoder models only the within-class residual, $\tilde{h} = \mathrm{ReLU}(m_y + \mathrm{dec}(z, m_y))$, where m_y is the exact class mean and enters through a class-*shared* projection. There is no class-specific parameter, so Theorem 2 has nothing to act on. (ii) *Moment correction* (MC) rescales each class's samples to the exact class mean and spread; in idealised form it is the minimum-displacement map with these moments (Proposition 11); after the final ReLU and with plug-in moments it matches them only approximately. (iii) *Gate.* A mean anchor cannot separate classes whose means are close (Proposition 10); the server computes the separation $F = \overline{\|m_c - \bar{m}\|^2} / \overline{V_c}$ from the same sums and, if $F < 0.05$, uses the original generator with HWA instead. Measured F is 0.005–0.013 on SVHN, 0.054–0.080 on CIFAR-10 and ≥ 0.18 elsewhere.

Heads and readout. Heads use logit adjustment with the client's own prior and *consensus soft labels* (CSL) on synthetic features, $t(\tilde{h}) = \frac{1}{2} e_y + \frac{1}{2} \frac{1}{G} \sum_g \mathrm{softmax}\, h_g(\tilde{h})$, the averaged prediction of all architectures' heads on that feature [27]. After training, every architecture's last layer is replaced by ER. *Optional:* kernel herding toward the exact class mean embedding adds only $+0.1$ to $+0.4$ points with ER at about $6\times$ the head-stage cost (Appendix C); it is not part of the method, but the fresh-seed runs below include it. The *minimal variant* for an existing deployment is the original generator with HWA (or decoupled decay), LA, CSL and ER.

6 Experiments

Protocol. We follow GeFL-F's published configuration [1]: a one-block frozen extractor, ten CNN head architectures, CVAE-F, SGD for the heads. Datasets: MNIST, FashionMNIST, SVHN, CIFAR-10 and CIFAR-100-LT; imbalance factor 100, Dirichlet(0.5) client shares, $K = 10$ (one client per architecture) unless stated. On our reproduction GeFL-F matches the published IID numbers within 0.5 points (MNIST 96.02 vs 95.47; FashionMNIST 82.67 vs 83.14). Metric: final-round balanced accuracy on balanced test sets. *Development* results (seeds 0–2) chose T_s, β, τ and the gate threshold on the test sets; the *main comparison* (Section 6.2) uses fresh seeds 3–5, chooses every method's

Table 2: Every method under the same readout (development seeds 0–2, long tail). The exact readout lifts GeFL-F by 7–20 points; with the same readout, our bodies are 2.0–3.9 points better. Brackets: 95% CI.

	MNIST	FMNIST	SVHN	CIFAR-10	CIFAR-100
GeFL-F, heads as trained	74.5	59.2	49.6	33.9	7.8
GeFL-F + ER	90.4	79.3	64.7	51.3	15.2
Minimal variant + ER	92.4	80.6	66.4	53.4	16.5
Ours, heads as trained	95.1	80.1	65.4	45.7	15.4
Ours + ER	**94.4**	**81.5**	**66.7**	**53.9**	**18.6**
Ours − GeFL-F, both + ER	+3.9 [3.0, 4.8]	+2.2 [0.1, 4.4]	+2.0 [0.8, 3.2]	+2.6 [2.1, 3.1]	+3.4 [3.1, 3.8]

Table 3: Main comparison on fresh seeds 3–5, every method's readout (and our T_s) selected on validation data. Ours minus baseline, test balanced accuracy, paired over seeds; brackets: 95% CI; bold: Holm-adjusted $p < 0.05$ over nine baselines. The four real-data baselines with LA, FedLC, FedRS and balanced softmax coincide (Proposition 5); we show two.

	MNIST	FMNIST	SVHN	CIFAR-10	CIFAR-100
Ours (test accuracy)	95.6	82.1	65.6	53.1	18.7
− GeFL-F	**+3.4** [3.2, 3.6]	+1.0 [−0.0, 2.1]	+1.5 [0.6, 2.5]	**+2.4** [1.8, 3.0]	**+4.0** [2.6, 5.4]
− GeFL-F, AdamW + LA	**+2.0** [1.9, 2.2]	+0.5 [−0.2, 1.2]	+1.5 [0.4, 2.6]	+1.5 [0.8, 2.2]	**+3.0** [2.3, 3.7]
− real-data heads + LA	**+4.7**	**+2.6**	**+11.3**	+1.5	**+1.6** [0.7, 2.6]
− real-data heads + FedRS	**+4.7**	**+2.6**	**+11.2**	+1.6	**+1.8**
− Gaussian generator + LA	**+2.9** [1.7, 4.1]	**+1.3** [1.3, 1.4]	**+12.4**	+0.5 [−1.1, 2.1]	**+1.6** [0.4, 2.8]
− FedDF	**+1.9** [1.4, 2.5]	+0.3 [−0.6, 1.2]	+0.5 [−1.0, 2.0]	+2.3 [0.9, 3.7]	**+3.5** [2.5, 4.5]
− FedDF + LA	**+2.0** [1.2, 2.8]	+0.3 [−0.4, 1.0]	−0.3 [−1.6, 1.0]	+1.7 [0.4, 3.0]	**+3.5** [2.2, 4.9]
margin over the strongest	**+1.9**	+0.3	−0.3	+0.5	**+1.6**

readout and our T_s on 2,000 held-out training images, fixed its selection rule and predictions before any run, and reports paired 95% CIs with Holm's correction. Details are in Appendix B.

6.1 Where the long-tail loss sits

Table 2: most of GeFL-F's long-tail loss is in the last layer (+15.9, +20.1, +15.1, +17.4, +7.4 points from ER). Against a recalibrated GeFL-F, the gain of a better body is 2–4 points under every all-client readout (CCVR +2.1 to +3.6; a pooled-real refit of the last layer +2.6 to +4.5). Published per-architecture readouts use only an architecture's own clients and lose 4–26 points against their all-client versions.

6.2 Fair re-test against nine baselines

Baselines (Table 3): GeFL-F; GeFL-F with decoupled decay and LA; heads trained on real data only with LA, FedLC, FedRS or balanced softmax; a Gaussian generator built from the same exact statistics, with LA (CCVR-style virtual features during training); and server-side ensemble distillation (FedDF [4]) on generated features, with and without LA. Each gets its own validation-selected readout. **Ours is not ahead of every baseline:** the margin over the strongest is −0.3 to +1.9 points, significant on MNIST and CIFAR-100 only. Our pre-registered prediction of at least 2 points over every baseline on four datasets *failed*. FedDF is strong once it gets an exact readout: distilling the ensemble into each head carries the ensemble's information even where the generator's labels carry none. Removing the coupled decay and adding ER recovers most of GeFL-F's loss; the anchored generator adds 0.5–3.0 on top, and on CIFAR-100 it turns GeFL-F's harmful synthetic phase (14.8 vs 16.9 for no synthetic data) into a gain (+1.6). Ablations are in Appendix C.

Table 4: Long-tail loss of the federated pipeline vs the centralised comparator (pooled real data, same frozen-extractor design, same readout). CIFAR-100: three of the ten architectures; one seed.

	MNIST	FashionMNIST	CIFAR-100
Body trained on one client's real data + ER	91.5	78.1	15.1
Ours + ER	94.4	81.1	18.7
Pooled body + ER, federated extractor	94.4	81.9	18.1
Long-tail drop, ours (balanced → long tail)	2.9	3.3	2.7
Long-tail drop, centralised pipeline	3.2	2.7	4.1
Federated residual (centralised − ours)	0.0	1.9	−0.6

Table 5: Robustness (balanced accuracy with ER; fresh seeds 3–5 unless stated). Left: several clients per architecture. Right: ours minus the decay-fixed GeFL-F under other imbalance settings, FashionMNIST / SVHN.

	SVHN (seeds 0–2)		CIFAR-10
K	50	100	50
GeFL-F	58.0	58.8	–
AdamW + LA	60.3	59.9	45.1
HWA + LA	60.3	60.8	45.2
Ours	**61.8**	**61.0**	**46.1**

setting	ours − decay fix
IF = 10	+1.0 / +2.6
Dirichlet 0.1	+1.5 / +1.7
Dirichlet 1.0	+0.8 / +1.0
random rare classes	+0.6 / +1.1

6.3 Federated versus centralised training

Table 4: a body trained on one client loses 3.0–3.8 points against the pooled body; our synthetic data recover all of it on MNIST and 80% on FashionMNIST, and on CIFAR-100 our bodies reach the pooled level (± 1). Our federated pipeline loses about as much to the long tail as the centralised pipeline does: *what remains is largely the long tail itself, not federation.* FashionMNIST's 1.9-point residual is about 1.1 extractor and 0.8 body. This also bounds what a better generator can still add over the decay-fixed baseline: if the pooled body is the ceiling, at most 1.9 points on MNIST (we measure $+2.0$) and 1.4 on FashionMNIST (we measure $+0.5$). The small margins of Table 3 are therefore expected, not a failure of tuning. SVHN and CIFAR-10 were not measured.

6.4 More clients and other imbalance settings

Table 5. With several clients per architecture, HWA's advantage over decoupled decay is $+0.9$ with ER on SVHN at $K = 100$ but about zero on CIFAR-10 at $K = 50$ ($+0.1$, CI $[-0.2, 0.4]$); ours is $+1.0$ ($[0.2, 1.8]$), and every method loses 6.5–7 points to $K = 10$. Under other imbalance settings and with rare classes drawn at random, ours stays ahead of both baselines in all eight cells, with a larger margin under stronger label skew (more absent classes leave more room for synthetic data, Proposition 7). The margins are small (three seeds).

6.5 The same collapse in federated recommendation

Section 3 needs only parameter rows used by few samples, clients that never use them, local Adam with coupled L2, and plain averaging. Federated matrix factorisation has all four [39]: item embeddings are shared and averaged, user embeddings stay local. On MovieLens-1M [40] (explicit ratings, users assigned at random to $K = 10$ or 50 clients, $d = 32$, 30 rounds, three seeds; predictions fixed before the runs), $\kappa_c < 1$ for 895 of 3,706 items at $K = 10$ and 2,108 at $K = 50$.

Table 6: κ_c ranks the collapse (Spearman with the final norm 0.93 / 0.96), the collapsed share of all items grows with K (22% to 48%), and it costs accuracy on these items. Two of our four predictions failed: not every $\kappa_c < 1$ item collapses (83–85% do), and *centralised* Adam with coupled L2 collapses 78% of them, because they receive data less often than once per 1,000 steps (Section 3). The share of optimiser steps whose batch contains the item predicts the final norm in both regimes (Spearman 0.91 central, 0.94–0.97 federated), and at $K = 50$ federation collapses moderately rare items about ten times harder. In recommendation, federation adds severity rather than the existence of

Table 6: Rare-item collapse in federated matrix factorisation (MovieLens-1M). Items with $\kappa_c < 1$; collapsed: final norm ≤ 0.15 of initial.

	collapsed $K = 10$ / 50	RMSE on these items $K = 10$ / 50	RMSE overall $K = 10$ / 50
Adam + coupled L2, FedAvg	85.0% / 83.4%	1.094 / 1.064	0.859 / 0.908
AdamW, FedAvg	0% / 0%	1.036 / 0.992	0.904 / 0.907
Adam + L2, HWA	23.3% / 8.7%	1.042 / **0.967**	**0.857 / 0.894**
SGD + L2, FedAvg	0% / 0%	1.160 / 1.144	0.967 / 1.010
centralised Adam + coupled L2	78.0%	1.099	0.888

collapse. Count-weighted item aggregation is known [41] and degraded item embeddings in federated recommendation have been reported [42]; we are not aware of an account of the optimiser mechanism or of a count-based predictor.

What did not work. A server-side signal-fraction gate, a class-balanced extractor loss and recomputing the exact readout during training did not help our method (Appendix E).

7 Related work

Model-heterogeneous FL. Methods exchange predictions on public or generated data (FedMD [3], FedDF [4], FedGKT [5]), keep part of the network local (LG-FedAvg [6], FedRep [7], FedProto [8]), train a server-side generator against the client ensemble (FedGen [9], FedFTG [10], DENSE [11]), share statistics (FedVTC [13], Felo/Velo [12]), or train the generator itself by FedAvg (GeFL [1]). FedGen and FedFTG weight teacher *losses* by label counts; HWA weights the *parameters* of a client-trained generator, and our contribution is the mechanism behind it.

Federated learning under label skew and long tails. Classifier re-calibration from federated statistics (CCVR [14], CReFF [15], Fed3R [22], FedNCM [23]) and loss corrections by client label frequencies (FedLC [16], FedRS [17], FedRoD [21], BalanceFL [18], Fed-GraB [19], FedLoGe [20]) are established. Proposition 5 explains why the loss corrections coincide for a single-client head, and Theorem 6 why the all-client readouts recover the last layer exactly. Class embeddings can collapse under FedAvg when clients see only positive labels [24], a different cause.

Long tails, optimisers and synthetic data. We build on logit adjustment [26], effective numbers [25] and decoupled classifiers [28]; the collapse rests on coupled versus decoupled decay in Adam [29, 30]. Conditional generators can collapse on limited classes centrally [31], and synthetic training data are limited by label fidelity [32].

8 Limitations and conclusion

Limitations. Our margins over strong baselines are small; the claim is the decomposition, not a new state of the art. The extractor is small and frozen, as in GeFL-F. The centralised comparison covers three datasets with one seed, and only the fresh-seed comparison chose settings without test data. The theory uses models, so we draw only ranking and sign claims from it, each paired with a measurement. The released class sums leak membership for rare classes unless noised (Appendix D). The recommendation result rests on one dataset.

Conclusion. Heterogeneous FL loses rare classes in three places: the optimiser's treatment of clients that lack a class, a last layer fitted to one client's prior, and synthetic data without label information. Each has an exact description and a direct fix; what remains is largely the cost of the long tail itself.

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A Proofs

Lemma 1. Given the counts, each contiguous chunk of a uniformly random permutation of a class's indices is a uniformly random subset; the step that guarantees a minimum client size moves uniformly random samples and preserves exchangeability within a class. Since ϕ is shared, $h = \phi(x)$ inherits the property. □

Stage I. The one-round expansion of local SGD gives the aggregate update $-\eta\,\bar{\tau}\nabla\tilde{\mathscr{L}} + O(\eta^2 LG\tau^2)$ with $\tilde{\mathscr{L}} = \sum_k \frac{p_k\tau_k}{\bar{\tau}}\mathscr{L}_k$ [35]. With $p_k = 1/K$ and $\tau_k = \lfloor 5n_k/B \rfloor$ (batches of size one skipped) the weights are n_k/N up to the rounding, whose total variation we computed on every split used. □

Theorem 2. $\|\frac{1}{K}\sum_k r_{k,c}\| \le \frac{1}{K}\sum_k \|r_{k,c}\| = \|r_c\| + \frac{1}{K}\big(\sum_{k\in H_c}\Delta_k + \sum_{k\notin H_c}\Delta_k\big)$; the two sums are $m_c\bar{G}_c$ and $-(K-m_c)\bar{D}_c$. Equality in the triangle inequality holds iff all vectors are non-negative multiples of one another. □

Lemma 3. A non-holder's batches contain no sample of class c, so the data loss does not depend on r_c and the coupled-decay gradient is λr_c. Adam's update $\eta\hat{m}/(\sqrt{\hat{v}}+\varepsilon)$ is unchanged when the whole gradient sequence is multiplied by $a > 0$, up to the ε term; hence the step does not depend on λ while $\lambda|r_{c,j}| \gg \varepsilon$. Numerically the invariance holds for $\lambda \gtrsim 10^{-5}$ at the measured row norms. Each client keeps its Adam state across rounds while its parameters are reset to the global ones, so a single step can exceed η. □

Proposition 5. (1) $f + a_k = Wu(h) + (b + a_k)$, and $b \mapsto b + a_k$ is a translation with identity Jacobian, so gradients agree at corresponding points; induct over steps. (2) $\alpha f_c = \tilde{w}_c^\top u + \tilde{b}_c$; the gradient in $\tilde{w}_c$ is $1/\alpha$ times that in w_c, so a step of size η on w_c is a step of $\eta\alpha^2$ on $\tilde{w}_c$, and the body sees the same logits. Numerically the identity holds to 4×10^{-16}. □

Theorem 6. Addition is associative and commutative. In particular every exponential-family maximum-likelihood estimate, and the class-balanced LDA of ER, is computed exactly by one round of secure aggregation. □

Proposition 7. One round is the affine map $\theta \mapsto P_r P_s\theta + P_r(I-P_s)\theta_s + (I-P_r)\theta_r$. If $\|P_r P_s\| < 1$ it is a contraction with fixed point $\theta^\star = (I - P_r P_s)^{-1}[P_r(I-P_s)\theta_s + (I-P_r)\theta_r]$; writing $(I-P_r) = (I-P_r P_s) - P_r(I-P_s)$ gives the stated form. With commuting Hessians all matrices are diagonal in a common eigenbasis. □

Proposition 8. $\frac{1}{C}\sum_c(\mathrm{LSE}(f) - f_c) = \mathrm{LSE}(f) - \bar{f}$ and $\mathrm{KL}(u\|\mathrm{softmax} f) = -\log C + \mathrm{LSE}(f) - \bar{f}$; expanding LSE around constant logits gives the quadratic term. The mixture form follows by conditioning on whether a sample is faithful. □

Proposition 9 (dilution without decay; quadratic row model)**.** *Let client k's class-c loss be quadratic, $\frac{1}{2}\sum_{i\in D_{kc}}\|r - t_i\|_A^2$. A local round then maps r to $M_{kc}r + (I - M_{kc})\bar{t}_{kc}$, and flat FedAvg converges to*

$$r^* = \Big(\sum\nolimits_{H_c}(I - M_{kc})\Big)^{-1}\sum\nolimits_{H_c}(I - M_{kc})\bar{t}_{kc}$$

at a rate diluted by m_c/K. With saturated local training ($M_{kc} \to 0$) its mean squared error is $V_c = \frac{n_c}{m_c^2}\sum_{k\in H_c} 1/n_{kc} \ge 1$ times that of the pooled estimate, with equality iff all holders have equal counts; HWA with weights $E(n_{kc})$ gives $V_c \approx 1$.

Proof. Non-holders contribute the identity map and drop out of the fixed-point equation; the rest is the fixed-point formula for quadratic clients and the variance of independent unbiased local means (Lemma 1). On the actual MNIST counts simulated error ratios match V_c to two decimals. □

Proposition 10 (what a mean anchor can express)**.** *Let the class-c sampler be* $X_c = \mathrm{ReLU}(m_c + g(Z, Am_c))$ *with Z shared across classes and g L-Lipschitz in its second argument. Then* $W_1(G_c, G_{c'}) \le (1 + L\|A\|_2)\|m_c - m_{c'}\|$.

Proof. Couple both classes through the same Z; ReLU is 1-Lipschitz. □

In units of the within-class spread, the anchors are on average $\sqrt{2F}$ apart. On SVHN ($F \approx 0.005$–0.013) they differ by 10–15% of the spread, which is why the gate of Section 5 falls back to class rows there.

Proposition 11 (minimum-displacement moment correction)**.** *Among all maps T with* $\mathbb{E}\,T(X) = m_c$ *and* $\mathbb{E}\|T(X) - m_c\|^2 = V_c$*, where X has mean* $\tilde{m}_c$ *and spread* $\tilde{V}_c$*, the map* $T^\star(x) = m_c + \sqrt{V_c/\tilde{V}_c}\,(x - \tilde{m}_c)$ *minimises* $\mathbb{E}\|T(X) - X\|^2$*. With exact covariances instead of spreads the minimiser is the linear map attaining the Gelbrich bound [43].*

Proof. With $Y = T(X)$, $\mathbb{E}\|Y - X\|^2 = V_c + \tilde{V}_c + \|m_c - \tilde{m}_c\|^2 - 2\mathbb{E}\langle Y - m_c, X - \tilde{m}_c\rangle$, and Cauchy–Schwarz bounds the last expectation by $\sqrt{V_c\tilde{V}_c}$, with equality iff $Y - m_c = \sqrt{V_c/\tilde{V}_c}(X - \tilde{m}_c)$. □

The deployed map applies a final ReLU and uses moments estimated from a finite batch, so it matches the moments only approximately.

B Experimental details

Models and optimisation. Extractor, heads (CNN-1 to CNN-10) and CVAE-F as in GeFL-F's appendix; CIFAR extractors have ten output channels. Heads: SGD, learning rate 0.1. Rounds for extractor / generator / heads 20/100/50 on MNIST and FashionMNIST, and as published elsewhere. Data fraction 0.1 (0.5 for CIFAR-10, 0.22 for CIFAR-100, which gives the standard CIFAR-100-LT shape: 480 images in the largest class and 5 in the smallest). Client-to-architecture assignment is contiguous. With IF = 100 the rarest classes are held by 6–8 of 10 clients; CIFAR-10's realised imbalance is 48:1 because the rarest classes are capped by the data fraction.

Selection and statistics. Fresh-seed runs choose every method's readout among heads as trained, balanced bias calibration, ER, Fed3R, FedNCM and CCVR (all-client or own-client) by validation balanced accuracy, and our $T_s \in \{1, 10\}$ the same way. Each seed fixes the split and the initialisations for every method; GPU training is not bit-deterministic (a repeated seed differs by -0.06 ± 0.43 points with ER over 21 paired runs). CIs are paired t-intervals over seeds.

The balanced setting. With IID clients ours is within $+0.1$ to $+1.6$ points of GeFL-F with ER (MNIST 97.4 vs 96.9, FashionMNIST 84.8 vs 84.5, CIFAR-10 61.6 vs 61.4, SVHN 77.3 vs 76.2, CIFAR-100 21.4 vs 19.8); the gain is significant on SVHN and CIFAR-100 only. Holder-weighted averaging equals FedAvg when all clients have equal counts.

C Ablation and mechanism controls

Five seeds, heads as trained: removing any component lowers MNIST accuracy in every seed (LA -3.05, mean anchor -3.89, CSL -1.35, MC -1.13, KH -0.67); on FashionMNIST LA -5.78, CSL -1.50, KH -0.65 (5 of 5), MC -0.63 and mean anchor -0.50 (4 of 5). With ER: LA $+0.11$ / $+0.05$ (its contribution vanishes, as Proposition 5 requires), CSL -0.44 / -0.53, MC -0.53 / -0.14, KH -0.26 / -0.13 (MNIST / FashionMNIST). Controls with one shared generator per seed: herding beats a random subset of the same candidates (small but consistent); MC's gain comes from the spread, not the mean (mean-only correction loses 0.9–1.0 points although it raises the referee's tail fidelity by 4–5 points); results are flat in the random-feature dimension and the candidate-pool size.

Kernel herding is not worth its cost. Removing herding, paired by seed, costs 0.26 (MNIST), 0.13 and 0.23 (FashionMNIST) and 0.37 (CIFAR-10) points with ER, and 0.2–0.7 as trained; replacing

it by a random subset of the same candidates costs 0.16–0.23 with ER. All 8 paired seeds with ER favour herding (sign test $p = 0.008$), but it makes the head stage 5.8–6.8$\times$ slower, and a larger random-feature dimension or candidate pool changes nothing, so a better approximation of the same kernel would not help. We therefore treat it as optional. The fresh-seed comparison of Table 3 includes it; without it our margins would be about 0.1–0.4 points lower (measured on MNIST, FashionMNIST and CIFAR-10).

D Privacy and cost

Table 7: Membership inference on the released class sums (AUC; all classes / tail classes / rarest class).

release	MNIST	FashionMNIST
exact sums	0.52 / 0.64 / 0.69	0.52 / 0.63 / 0.73
Gaussian mechanism, $\varepsilon = 8$	0.50 / 0.58 / 0.64	0.52 / 0.54 / 0.45
Gaussian mechanism, $\varepsilon = 2$	0.50 / 0.53 / 0.61	0.51 / 0.50 / 0.41

An attacker who knows a target's label and holds public reference data scores it by how far the released class mean moved toward it. Head classes are at chance; the exact release leaks membership for rare classes (24–112 samples), as expected for a mean over few samples. Formal differential privacy on the four statistics our generator adds costs 4.5–6.9 points at $\varepsilon = 8$ and 9.8–14.1 at $\varepsilon = 2$, still above GeFL-F at $\varepsilon = 2$ on MNIST and FashionMNIST; it does not cover the generator's FedAvg training or the heads. Our pipeline is 2.2–2.8$\times$ slower than GeFL-F end to end with the optional herding, which dominates the head stage; ER adds one round and up to about 5×10^5 numbers per architecture per client.

E Further negative results

A server-side estimate of the signal fraction used as a gate helps the decay-fixed GeFL-F on the CIFARs ($+0.8$, $+1.3$) but not ours, because its LDA-based estimate saturates under moment correction (Spearman correlation with the referee's signal fraction 0.48). Weighting the extractor's loss to the balanced objective was neutral on MNIST and FashionMNIST and harmful on the CIFARs (-8.2, -4.2), where the unnormalised weights destabilised training.

Recomputing the exact readout every round and training only the bodies under it: 93.9 / 79.4 (MNIST / FashionMNIST) against 94.1 / 79.8 for ER computed once, with much worse calibration. Server-side distillation of the ensemble without an exact readout lowered IID accuracy. A Gaussian generator from exact statistics and a hybrid of anchor and class rows were weaker than the gated generator.